Automatic Question Answer Generation Based on Gujarati Paragraph

Mid-Sem Project Presentation

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Presentation Outline

1. Outline
2. Problem Statement and Contributions
3. Datasets
4. Overall Workflow
5. Classification Methodology
6. Classification Results
7. Generation – Models Explored
8. Why mT5?
9. mT5 Training Details
10. Sample Outputs
11. Conclusion and Future Work

Problem Statement

- Automatic Question Generation (AQG) for Gujarati is challenging because Gujarati is a low-resource language and high-quality paragraph-level QA datasets are limited.
- The target is not only to generate a single factual question, but to eventually generate multiple grounded questions covering the paragraph content and Bloom's taxonomy levels.A practical system must generate questions only from the paragraph, avoid hallucination, and produce useful answerable QA pairs.

Current Mid-Sem Focus

1. **Bloom's taxonomy question classification** using Gujarati questions.
2. **Answer-aware Gujarati question generation** using context + answer as input.

Key Mid-Sem Contribution

- Built Gujarati datasets for classification and generation.Implemented multiple generation approaches: template, mT5, IndicBART, BiLSTM baseline, and hybrid reranking pipeline. Identified the main research bottleneck: **dataset design for paragraph coverage and question diversity**.

Datasets for Two Tasks

Classification Dataset

- 900 Gujarati questions.
- 9 Bloom types, 100 each.
- Balanced, cleaned, deduplicated.

Example: “ભારતની રાજધાની શું છે?” □ Factual

Generation Dataset

- 1000 short literature paragraphs.
- One QA pair per paragraph.
- Answers are short phrases (grounded).

Example:
Context: કલાપી ... કવિ હતા ...
Q: કલાપીનું મૂળ નામ શું હતું?
A: સુરસિંહજી તખ્તસિંહ ગોહિલ.

Two-Stage System Workflow

Classification

900-question Bloom dataset

↓

TF-IDF (n-grams 1-3)

↓

Logistic Regression

↓

99% accuracy, confusion matrix

Generation

1000-pair literature dataset

↓

Six models tested

↓

mT5-small (best)

↓

BLEU 0.34, ROUGE-L 0.42

Classification – TF-IDF + Logistic Regression

- **Preprocessing:** clean text, keep only Gujarati/Devanagari chars.
- **Feature extraction:** TF-IDF with character n-grams (1–3).

$$\text{tfidf}(t, d) = \text{tf}(t, d) \times \log\left(\frac{N}{\text{df}(t)} + 1\right)$$

- **Classifier:** Multinomial logistic regression, softmax output.

$$P(y = c|\mathbf{x}) = \frac{\exp(\mathbf{w}_c^T \mathbf{x} + b_c)}{\sum_j \exp(\mathbf{w}_j^T \mathbf{x} + b_j)}$$

- **Optimization:** cross-entropy loss, solver `saga`, $C = 4.0$.

Classification Results

Metrics

Test accuracy	98.9%
10-fold CV accuracy	99.6%
Macro F1	0.99

Misclassifications (only 2)

1. મોંઘવારીમાં ... ભૂલો થાય છે? (True: Conceptual, Pred: Compare)
2. પૃથ્વીનો વ્યાસ કેટલો છે? (True: Factual, Pred: Compare)

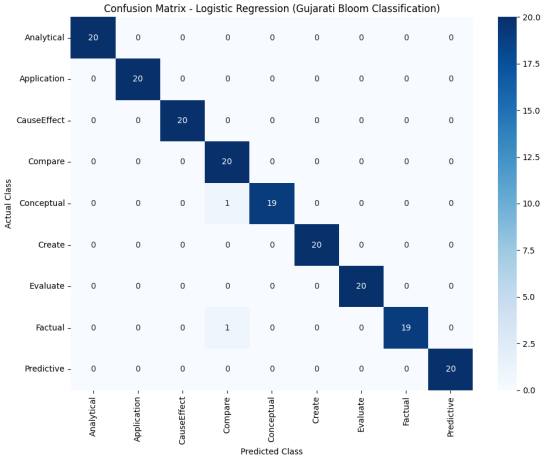


Figure: Confusion matrix – 9 classes.

Generative Models – Systematic Comparison

Model	**BLEU**	**ROUGE-L**
Template-based	0.21	0.38
Seq2Seq (LSTM)	0.08	0.18
Bi-LSTM + attention	0.09	0.20
GRU Seq2Seq	0.07	0.16
mT5-small	**0.34**	**0.42**
IndicBART (80 steps)	0.22	0.31

- mT5-small outperforms all others.
- 30% of outputs still contain `<extra_id_...>` → under-training due to limited data.

Generation Approaches Implemented

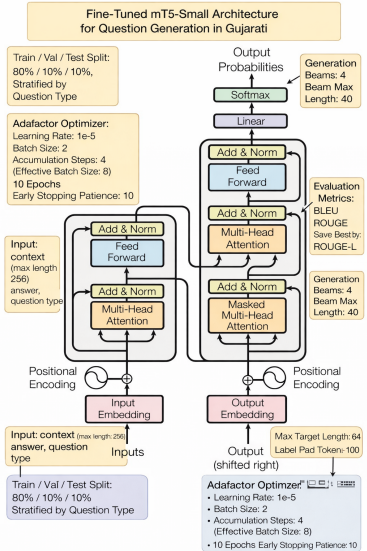
Approach	Purpose in the system
Template-based baseline	Simple rule pattern using answer type
Smart rule-based template	Better Gujarati question patterns from keywords in context/answer
mT5-small	Main transformer-based answer-aware question generator
IndicBART	Indic-language seq2seq alternative for Gujarati generation
BiLSTM baseline	Neural baseline to study sequence learning behavior
Hybrid pipeline	Candidate generation + reranking + fallback selection

Mid-Sem Observation

The implemented approaches show that transformer-based models are more promising than shallow baselines, but the current dataset is still too limited for stable paragraph-level question generation.

Why mT5-small Was Chosen

- Multilingual pre-training on 101 languages $\rightarrow$ strong transfer.
- Text-to-text format aligns perfectly with our formulation.
- Encoder-decoder with attention uses both context and answer.
- More stable than IndicBART in our experiments.



mT5-small Training Hyperparameters

Parameter	**Value**
Learning rate	1×10^{-5}
Batch size (effective)	8
Gradient accumulation	4 steps
Max input length	256
Max target length	64
Optimizer	Adafactor
Epochs	10
Beam size (inference)	4
Early stopping patience	10

How mT5 Learns During Training

Core Learning Logic

For each training pair, the model tries to maximize the probability of the correct question tokens:

$$P(y|x) = \prod_{t=1}^{T} P(y_t \mid y_{<t}, x)$$

where:

- x = input text (context + answer) y_t = target token at step t

Gujarati Example

Input: context + answer = જ્ઞાનપીઠ પુરસ્કાર Target question tokens: ઉમાશંકર | જોશીને | કયો | પ્રતિષ્ઠિત | પુરસ્કાર | મળ્યો?

- At each decoding step, the model predicts the next token. If the predicted token is wrong, loss increases. Through repeated updates, the model gradually learns question patterns.

Sample Generation from mT5-small

Correct Example

Input: `generate question in Gujarati: context:` કલાપી (સુરસિંહજી તખ્તસિંહ ગોહિલ) ગુજરાતી ભાષાના પ્રખ્યાત કવિ હતા... `answer:` સુરસિંહજી તખ્તસિંહ ગોહિલ `question type: factual`
Output: કલાપીનું મૂળ નામ શું હતું?

Still Under Refinement

About 30% of outputs contain `<extra_id_...>` or are incomplete – indicates need for more data / longer training.

mT5 Approach and Why It Was Used

- mT5 is a multilingual **text-to-text transformer**, so it naturally fits the task:

$$\text{context} + \text{answer} \rightarrow \text{question}$$

- It uses subword tokenization, which is helpful for Gujarati because rare words can be split into smaller meaningful units.
- It is pre-trained on multilingual text, so fine-tuning can transfer general language knowledge to Gujarati AQG.

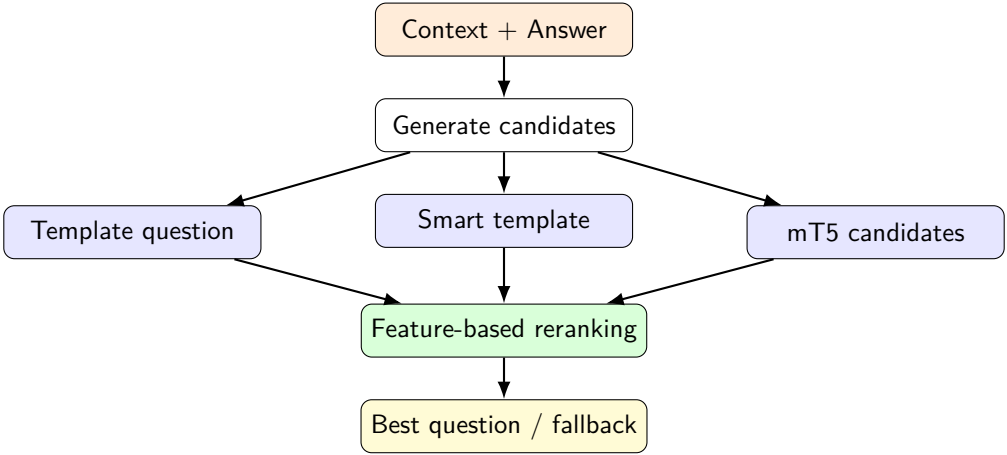
What the model learns

Given the paragraph and answer, the encoder reads the input sequence and the decoder generates the question token by token.

Observed Mid-Sem Result

The model architecture is suitable, but the current dataset is too small and too narrow (mostly one factual question per paragraph), so generation quality is still unstable.

Hybrid Generation and Reranking Pipeline



Why this helped

When raw neural generation became noisy, the hybrid pipeline could still use template-based candidates and select the most reasonable output through reranking.

Reranking Features Used

Feature	Meaning
`q_len`	Number of words in the candidate question
`a_in_q`	Whether the answer appears directly in the question
`has_question_mark`	Whether the candidate looks like a question
`context_overlap`	Number of overlapping words between context and question
`answer_len`	Length of the answer phrase

Model Used

The overall pipeline combines **mT5-based candidate generation** with a **Logistic Regression reranker**. The reranker scores each candidate question using handcrafted features and selects the highest-scoring output.

Important Limitation

These are shallow handcrafted features. Therefore, some semantically bad candidates may still get high scores if they look structurally similar to good questions.

mT5 Candidate Generation and Feature Extraction

Step 1: mT5 Generates Candidate Questions

Context: ઉમાશંકર જોશી ગુજરાતી સાહિત્યના પ્રગતશીલ કવિ હતા જેમને જ્ઞાનપીઠ પુરસ્કાર મળ્યો.
Answer: જ્ઞાનપીઠ પુરસ્કાર

Generated candidates:

- કયો પુરસ્કાર મળ્યો?
- ઉમાશંકર જોશી કયા માટે જાણીતા છે?
- ગુજરાતી સાહિત્યના કયા નાના કોણ છે?

Step 2: Feature Extraction

For each candidate, features are computed:

- q_len: number of words, a_in_q: answer present or not, has_question_mark: looks like question, context_overlap: common words with context

Logistic Regression Reranking

Step 3: Scoring Each Candidate

Each candidate is converted into a feature vector:

$$\mathbf{x} = [q_len, a_in_q, has_question_mark, context_overlap, answer_len]$$

The reranker computes:

$$\text{score} = \mathbf{w}^T\mathbf{x} + b$$

Interpretation

- Relevant and clear questions $\rightarrow$ higher score
- Noisy or unrelated questions $\rightarrow$ lower score

Example

કયો પુરસ્કાર મળ્યો? = High score ગુજરાતી સાહિત્યના કયા નાના કોણ છે? = Low score

Final Question Selection

Step 4: Select Best Candidate

The candidate with the highest score is selected as the final output.

Gujarati Example

Best selected question:

કયો પુરસ્કાર મળ્યો?

Key Idea

- mT5 $\rightarrow$ generates multiple possibilities
- Logistic Regression $\rightarrow$ selects the best one

Observed Outputs from Experiments

mT5 Example from Notebook

Context: ગોવર્ધનરામ ત્રિપાઠીની નવલકથા સરસ્વતીચંદ્ર ગુજરાતી સાહિત્યનાં શ્રેષ્ઠ સર્જનોમાંની એક ગણાય છે. Answer: સરસ્વતીચંદ્ર Actual Question: ગોવર્ધનરામ ત્રિપાઠીની શ્રેષ્ઠ નવલકથા કઈ છે? Predicted: ના□□□ ગુજરાતી સાહિત્યના કયા નાના કોણ છે?

Hybrid / Template-Assisted Example

Context: ઉમાશંકર જોશી ગુજરાતી સાહિત્યના પ્રગતશીલ કવિ અને વિવેચક હતા જેમને જ્ઞાનપીઠ પુરસ્કાર મળ્યો. Answer: જ્ઞાનપીઠ પુરસ્કાર Best candidate selected: કયો પુરસ્કાર મળ્યો?

Error Analysis and Key Findings

- **Repetition/noisy generation:** mT5 often repeats high-frequency words such as ગુજરાતી, સાહિત્ય, or કયા.
- Weak dataset supervision: most contexts contain only one factual question, so the model does not learn diversity.
- Poor paragraph coverage: current answer extraction is based only on short chunks, some important facts are missed.
- Shallow reranking: structurally valid but meaningless candidates may still receive high scores.
- BiLSTM limitation: current baseline predicts only the first token, not the full question.

Main Mid-Sem Insight

The main challenge is not only model architecture. The bigger issue is the mismatch between the final goal (multi-question paragraph coverage) and the current dataset (mostly one easy factual QA pair per paragraph).

Discussion

- **Classification module:** strong and near-complete, with high performance on Gujarati Bloom-type prediction.
- **Generation module:** full experimental pipeline has been implemented and tested.
- **Most important result:** current outputs show that model quality depends heavily on dataset design.

What worked

- Clean supervised formulation: context + answer $\rightarrow$ question
- **Roadmap:** Hybrid system – answer extraction + mT5 generation + RankSVM reranking + template fallback. Hybrid pipeline improves robustness

What still needs improvement

- Multiple QA pairs per paragraph
- Better answer extraction and semantic reranking
- Human evaluation for grounding and fluency

Conclusion and Next Steps

Mid-Sem Conclusion

- The classification task for Gujarati Bloom-type questions is well established.
- A complete Gujarati AQG experimental pipeline has been implemented using template, transformer, mt5 - small, and hybrid reranking approaches.
- Current experiments show that the main research bottleneck is **dataset richness**, not only model selection.

Next Steps

1. Expand each paragraph to multiple QA pairs instead of only one.
2. Improve answer extraction from paragraph content.
3. Fine-tune mT5 / IndicBART on richer and more diverse supervision.
4. Evaluate with grounding, fluency, answerability, and paragraph coverage.

Future Work

Mid-Sem Conclusion

- Classification module is complete and ready for deployment (99% accuracy).
- Generation workflow is established; mT5-small identified as best direction.

Immediate Future Work

1. Expand generation dataset to 10,000+ pairs.
2. Longer fine-tuning of mT5-small and IndicBART.
3. Implement hybrid pipeline with answer extraction and reranking.
4. Human evaluation of fluency, relevance, grounding.

Thank You!

Questions?

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